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METHOD STATEMENT

(25 Meter Radar Tower Erection/Installation)

Company Name:	Naresh and Co
Company Address:	718/3, TH Road, Thiruvottiyur, Chennai 600019

Responsible Person:	Sivam
Phone:	+919884056221
Email:	nareshandco@yahoo.com

RAMS Developed By | Name:	Antony Jebastine		
Phone:	+919840000800	**Email:**	Jebastine006@gmail.com

Project Title:	Erection of 55-ton 25-meter Lattice Towers
Site Address / Location:	Address 1: Royal Thai Air Force Wing 7, Maluan Sub-District, Phunphin District, Surat Thani Province 84130 Thailand. Address 2: Royal Thai Air Force Wing 4, 305 Phahonyothin Road, Takhli District, Nakhon Sawan Province 60140 Thailand.

Brief Description of Work to be Carried Out:

Assembling and erecting 2 numbers of 25 meters, 55 tones each steel lattice tower.

Assembling of 5 Sections involve focusing on adherence to engineering drawings, lifting each segment with the help of cares and positioning them on the foundation and assembling verticals one above the other without any deviation in angles as per the drawing.

After 4 Segments are lifted and positioned, dog hose shall be lifted and mounted at the appropriate position with cross frame support and bolts and nuts.

Segment 5 shall be positioned above the dog house and fixed. Post which the platforms, rails and ladder shall be Assembled, and tested for firmness.

All the nuts and bots shall be tested for required torque

True vertical shall be testing before handing over for active installation

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Specific Work Location:	Royal Thai Airforce wing 4 and Wing 7
Estimated No of Workers:	11 members
Estimated Duration of Works (Include Start / End Dates)	**15 days per location, Total 32 days to complete 2 locations including 2 days for shifting to next location** **17 Mar: Check into accommodation near Wing 7** **18 Mar to 1 Apr: Erection of Tower at Wing 7** **2 Apr: Check out of accommodation near Wing 7** **2-3 Apr: Travel to Wing 4** **3 Apr: Check into accommodation near Wing 4** **4-18 Apr: Erection of Tower at Wing 4** **19 Apr: Check out of accommodation near Wing 4**

Name of First Aid Responder:	Ramesh	Mobile Number:	Shall be provided once obtained local SIM
First Aid Box Stocked and Available:	Yes		
Location of First Aid Box:	Close to the erection site, shall be showcased with all team members working at site		
Address of Nearest Hospital / Doctor:	Surat Thani Hospital: 56 48F5+MJW หมู่ที่ 2 Makham Tia, Mueang Surat Thani District, Surat Thani 84000, Thailand		
Nearest Hospital / Doctor Phone Number:	Surat Thani Hospital, +66 77 952 900		

Scope of Work:

Assembling and erecting 25-meter 55 ton steel lattice tower in Thailand (Wing 7 and wing 4) This erection shall be done by assembling sections first on the ground and then erecting the section one above the other(Section Method) involving several critical steps, focusing on adherence to engineering drawings, Lifting, Alignment, quality control, and safety standards which is explained below:

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Manpower:

- Sub-Contractors 80 Ton heavy duty Mobile Crane Engineer/ operator
- Sub-Contractors 30 Ton Mobile crane erector
- Sub-Contractors Cherry picker operator
- QC / Site Supervisor
- Health and Safety office / First aider
- Riggers
- Helpers

S. N.	Position	Number of Employees
1	Quality Engineer	Ramesh
2	Project Manager	Sivam
3	Site Supervisor	Sivam
4	Site worker	Riggers
5	HSE Officer	Ramesh

Tools and Equipment's:

The below equipment's and tools need to be arranged by Easat on site

1. 80-ton capacity heavy duty mobile crane with 50 meters boom lifting height – 1 No
2. 30-ton capacity mobile crane with 40 meters boom length – 1 No
3. Crane Fitness certificates
4. Four no's of slings (Nylon/synthetic) (each 10 ton capacity) and 10 meters length
5. 20ton pulley with chain blocks and 2:1 advantage
6. Slings, wire ropes, hooks, hoists, turnbuckles, shackles, beams and spreader bars.
7. Theodolite or total station with operator to verify if required.
8. First aid box with standard medicines and bandage kits
9. Water cans, small water glass.
10. Hydraulic wrench for standard torque of all bolts
11. Barricade tapes (200 meters) Reflector cones, Men at work and danger sign boards
12. Accessories such as, MM Wire Rope, D-Shackle, Crowbar, Pulley Open and Close type, torque Wrench, Box Spanner, Hammer 5Kg, Spanners sets 3, Lifting Belt 4(NOS) (10T each)

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Pre-Erection Preparation and Material Inspection:

Firm concrete and even surface of 10 sq meter space to position the above crane's and anchor its extended arms with perfect balance.

Material Verification: All steel members, bolts, nuts, and washers will be checked against the Bill of Materials (BOM) and engineering drawings to ensure the correct parts are available at the site before the start of work.

Quality Check: Inspect materials for any damage, bent parts, excessive wear and tear or any Damage or undesirable components if any will be noted and reported

Foundation/ Base structure: Check and Ensuring the tower base concrete and base foundation of the structure is been properly grounded as per drawing and no deviation found.

HSE Requirements and Safety verifications:

1. Safety checks will be made before commencement of work to ensure that materials and equipment are in safe working order following project and site regulations
2. All personal will have had the site induction safety training.
3. All the riggers working on height shall be certified
4. All certificates and licenses of mobile cranes and crane operators will be checked before the reaction of tower cranes.
5. The erecting will be carried out strictly following the manufacturer's recommendations, a copy of which will be on site.
6. All riggers will be experienced and competent in the erecting of lattice tower.
7. All riggers have been trained and manually handled the same tower alignment
8. When working at heights above 2 meters lanyard body harness will be worn and used when exposed to open edges, harness to be attached to crane mast sections.
9. Hazard zone will be set up around the working area using safety barrier tape, together with danger signs to prevent unauthorized persons from entering the danger zone
10. Operations to take place only where weather and lighting conditions permit. The maximum wind speed for safe working shall be not greater than 40 Km/Hr.
11. Any circumstantial changes on site will be notified to the project managers..

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Safety and PPE Equipment

All necessary PPE and safety gear will be worn at all times during the execution of work, in full compliance with safety requirements as below.

- Helmets
- Safety Shoes
- Hand gloves
- Safety full body Harness
- High Visibility Vest

Roles and Responsibilities

Site Supervisor Cum Project manager:

- The carrying-out of work and the proper distribution of all the available resources in on daily basis
- Supervise and give an assignment for the team and coordinate with Crane operator
- Provision of all necessary information and distribution of responsibilities to his site team.
- The constant coordination with the Safety Engineer to ensure that the works are carried out in a safe working atmosphere.
- The constant coordination with the client for any works to be carried out and initiate for the Inspection for the finished works.
- He will ensure the implementation of any request that might be raised by the client at site.
- Efficient daily progress shall be obtained for all the equipment and manpower.
- He will engage in the work and check the same against the daily report received from the program manager.
- The passage of all the revised information to the supervisor and ensure that it's being carried out properly.
- Daily reports of the works are achieved and coordinated for future planning with the Site Engineer.
- Incorporate all the QA/QC and Safety requirements as requested by the concerned Engineer.
- Meeting with any type of unforeseen incident or requirement and reporting the same to the Site Engineer immediately.
- Conduct tool room meeting before start of work

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HSE Officer:

- Implementation of all safety measures following the HSE plan and that the whole workforce is aware of its proper implementation.
- Implementation of safety measures is adequate to maintain a safe working environment on the maintenance activity.
- Inspection of all the site activities and training personnel in accident prevention and its proper reporting to the Supervisor.
- The site is maintained in a clean and tidy manner.
- Ensure only trained persons shall operate the power tools.
- Ensure all concerned personals shall use PPE and all other items as required.
- Ensure high risk elevated areas are provided are barricades, tape, safety nets and provided with ladders.
- Ensure service area/inspection area openings are provided with barricades, tape, and safety nets.
- Provide necessary first aids and coordinate with nearby hospital and ambulance service where ever applicable

Riggers:

- Fixing all the angles, cross beams, main sections and preparing the segments.
- Will be climbing the towers to connect/disconnect and fix the segments
- Ensuring precise alignment of tower sections for bolting.
- Installing safety netting or temporary work platform
- Carefully working and positioning the segment for 100% alignment when the load is lowered by the crane
- Attaching lifting gear securely to tower components to maintain balance and stability during the lift.

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- Using clear, standardized hand signals or radio communication to direct crane operators during lifting and positioning.
- Guiding components into place, securing them, and, in many cases, bolting or assembling structural components at heights.
- Maintaining a clear work area, enforcing safety protocols, and ensuring structural integrity throughout the assembly process

Preliminaries for crane readiness:

The crane crew on-site shall familiarize themselves with site tasks to be undertaken.

- The Crane engineer shall ensure that the crane crew attends Site Safety Induction Training with the Main Contractor's Safety Officer before beginning the crane erection work.
- Components of the tower cranes shall be delivered to the site by a flat-bed trailer.
- Upon arrival at the site, the crane crew shall check and verify that delivered components are in proper working order and as commissioned.
- The components will be offloaded from the trailer/Container by the 30-ton mobile crane.
- The offloaded components should be taken and stored in the designated area.
- The crane operator shall work in coordination with the slingers/riggers to ensure safe lifting and slinging operations at all times.
- The supervising Engineer shall report any damaged or missing crane components before starting erecting. The crane crew shall prepare the crane for erecting.
- All listed operations below will be carried out by competent crane erectors and erecting crew working from crane mast, top of jib and counter jib. Erectors shall wear safety harnesses at all times, which shall be securely attached to the crane during height work.

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Slewing Ring and counter jib pre-assembly

- The crane crew shall assemble the slewing ring platform and cathead in the designated assembly area and prepare them for assembly to the tower crane
- The counter jib sections shall be prepared by the crane crew in the designated assembly area and assembled at ground level by use of the mobile crane.
- The crane crew shall connect the tie bars, on each side of the counter jib assembly.
- By use of pins.

Jib Assembly and Slewing Ring assembly to tower crane

- The crew shall prepare the site for 80-ton mobile crane. Once the crane is in place, the assembled slewing ring and cathead shall be lifted into place and fixed by experienced riggers
- The crane crew shall prepare the sections for the main jib in the designated assembly area.
- The crane crew shall carry out the main jib assembly at ground level, with wooden blocks under each assembled jib section, by use of crane
- The crane crew shall then assemble the tie bars, and attach them to the hooks on the assembled main jib.

Counter jib and jib assembly to crane

- Using the mobile crane, the counter jib assembly with the tie bars shall be lifted to the top of the cabin section and fixed to the above by pins provided.

Final assembly & Crane Construction Inspection

- When all work is completed, further mast sections shall be installed by using the telescoping cage, until the crane reaches the maximum freestanding height.
- Following completion of the assembly, full inspection and testing of the tower crane shall be carried out by a recognized 3rd party agency or inspection engineer.
- Any comments or discrepancies shall be rectified until the crane is deemed fit for use and the 3rd party or Quality engineer certificate can be issued.
- When all the above is satisfactory then the tower crane can be put in service.

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Foundation/ Base bolt Verification:

Tower base bolt verification is a critical process before the start of construction procedure to ensure the structural integrity, alignment perfection of the tower

It involves verifying that bolts are correctly installed, tightened to the specified torque, and free from corrosion with right measurements and 100% evenly positioned.

Below procedures will be followed and ensured.

- **Concrete Curing & Backfilling:**

  Confirm that the foundation concrete has reached sufficient strength, typical time frame should be of minimum 14 days after RCC. Ensure the foundation properly backfilled before beginning assembly.

- **Stub Alignment & Position:** Surveyors must verify the exact position of the foundation stubs, checking the back-to-back distances (both face and diagonal).

- **Verticality and Slope:** Check the verticality (plumb) and inclination/slope of the stubs is tested and certified by the surveyor.

- **Bolt/Stub Inspection:** Inspect all foundation bolts for proper size, damage, or degradation of threads.

- **Elevation Check:** Verify the relative levels of the top of the stubs to ensure a level base for the tower.

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Pre- Installation Preparation

Material Verification:

All steel members, grills, ladders, dog house, platform accessories, bolts, nuts, and washers will be checked against the Bill of Materials (BOM) and engineering drawings to ensure the all required materials are correct and available at the site before the start of work.

Visual Verification/ Quality Checks:

Inspect materials for any damage, bent parts, or excessive abrasion of material. Any such Damaged or undesirable components will be recorded and informed to Project Manager and to Client manager before using those.

Section Assembly (Sub-assemblies):

Correct Grouping: Tubes and members are separated and grouped by section (only if applicable) according to identification numbers and engineering drawings.

Avoid Force Fitting: Members will he fitted together without undue force. Using excessive force or cutting members to fit will be strictly prohibited which can compromise structural integrity.

Tower members will be "Assembled" (piece-by-piece) in a group for alignment adjustments required for segment preparation before the tower installations.

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All 5 segments (5 Meters each) will be aligned and installed side by side near (Appr 20 to 30 meters distance) by the actual tower erection location.
Once all the 5 Segments are assembled and kept ready, the required verification will be done for lifting the segment no 5 (The Bottom Most Segment) to ensure the slings, crane hooks and distance, load are all within the safety range

Stiffener Angle:

4 Temporary pieces of iron angles used will be fixed at all four sides of the segment at the bottom and all the 4 main legs of the segment will be connected to reduce the shear lag effect and increasing the efficiency of the tension member during lifting or assembly. to manage loads, stabilize members, or arrest movement (including limiting unwanted sag/flexure) during lifting.

Erection (Segment Method):

"Segment Built-up" method, larger segments (5 meters per segment) which are fully assembled on the ground already will be hoisted as a single unit which will be lifted and positioned starting from the bottom foundation upwards.

Lifting Configuration: 80 to mobile crane is positioned (1 no of 80 Ton heavy duty Mobile crane with 32 to 42.5 meters boom Length and lifting height up to 35 to 45 meters will be used for segment lifting and fixing

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1. **Lifting and Fitting – Segment 5:** The pre-assembled section 5will be lifted by the crane with the help of 2 to 4 Neylon or synthetic Slings of capacity 10 ton each with app 1-0 meters length.

   Rope from all 4 legs will be tied and the movement of the segment once lifted by the crane will be controlled until the segment is positioned on the foundation base bots correctly and secured by riggers at all 4 foundations.

   Necessary inclination of the legs shall be verified for the alignment of the segment,

   The stiffener angle to be removed after 100% segment is seated on bolted with the base bolts,

2. **Securing the Section:** The section is immediately bolted, often with two sides (opposite faces) lifted first and held in place by props or ropes, followed by the remaining sides, which are then laced together.

3. **Progressive Assembly:** This process is repeated, moving up the tower from bottom to top.

4. Similarly, segment, 4,3 and 2 shall be lifted, positioned and lowering and fixing lowered onto the stubs or the previously installed lower section with the help of care, riggers and supervisors.

5. **Dog House:**

Dog house shall be assembled on ground and then lifted with the help of crane, sling and hooks, will be positioned above the segment 2 at the right groove as per the drawing and to be secured**.**

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6. **Segment 1 and Platform:**

   Post the installation of dog house section 1 to be lifted and positioned as per the above process and procedures followed by the Platform and grills.

7. The ladder shall be either simultaneously along with the segments shall be installed or will be installed post dog house positioning.
8. **Bolting Procedure:** Correct size nuts and bolts with flat and spring washers as specified in the drawing will be used for tightening. Will Ensure a minimum of two threads project outside the nut after full tightening.
9. **Tightening Sequence:** Bolts will be tightened progressively from the top of the completed section downwards to ensure even stress distribution. A torque wrench may be used for final tightening checks.

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Alignment Checks:

10. Surveying tools (e.g., laser levels, theodolites, Total Station plumb bobs) to be used to verify the verticality and horizontal alignment of the structure at various stages, especially at connection points (nodes). Adjustments will be made to maintain specified tolerances. Engineer to operate the above tools to be there on ground for conducting this test from client side, If not plumb bob test shall be done for true vertical.

11. **Finalizing:** Cross-arms are generally assembled on the ground and lifted last, followed by final tightening, verticality checks, and bolt sleeves

Safety and Technical Requirements

- **Foundation Strength:** Erection should not begin until the concrete has cured for at least 14–21 days.
- **Handling:** Tower members should not be dragged to prevent damage to the galvanized coating.
- **Equipment:** Use of inspected, certified cranes, along with taglines to control the lifted load, is essential.

Completeness: Ensure all members and accessories are on place as per the drawing

Safety Precautions

Work will not be carried out during inclement weather (heavy wind, rain, thunderstorms, lightning).
Only experienced and medically fit personnel (tested for acrophobia) with valid working on height certification will be deployed to work at height.
All workers at height will be using EU standard full-body safety harnesses with double lanyards attached to secure anchor points along with industrial standard shoes and helmets.

Tools will be carried in a tool bag, and nothing will be dropped from height.
The work area will be barricaded to prevent unauthorized access. And minimum first aid kits will be readily available as per the health and safety standards.

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Out of Scope:
Tools and Machineries that needs to be arranged at site:

SL NO	Description	Qty.	Photo (example)
1	80 Tons Capacity Heavy Duty Mobile Crane with 50 Meters Boom Lifting Height	1	
2	Firm Concrete and even surface of 10 sq.m. space to position above crane and position its extended arms with perfect balance	1	To Winthstand crane load when lifting at the site. Idealy from 5 to 8 Meter away from Tower Location)

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3	30 Tons Capacity Mobile Crane with 40 Meters boom length	1	
4	Sling (Nylon/synthetic) 10 Tons capacity and 10 meters length	4	
5	Fork Lift (2 TON, 3 to 5 meter Lift capacity, Fork length 1000 to 1300 MM	1	
6	Slings (2 to 3 Tons)	2 NOS	
7	Wire Ropes (15 Ton Capacity) As part of Crane accessories	1	
8	Hooks (Part of crane accessories)	1	
9	Hydraulic wrinch for standard torque(19 to 45 MM Bolt Size) 3,000 - 6000 ft-lbs	1	
10	Shackles (Part of crane accessories)	2	

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11	Speader pars (2 Meter Width) 2 to 4 tonns capacity	1	
12	Theodolite or Total Station with operator to verify (if required)	1	
13	First aid box with standard medicines and bandage kits	1 Set	
14	Water cans, small water glass		

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